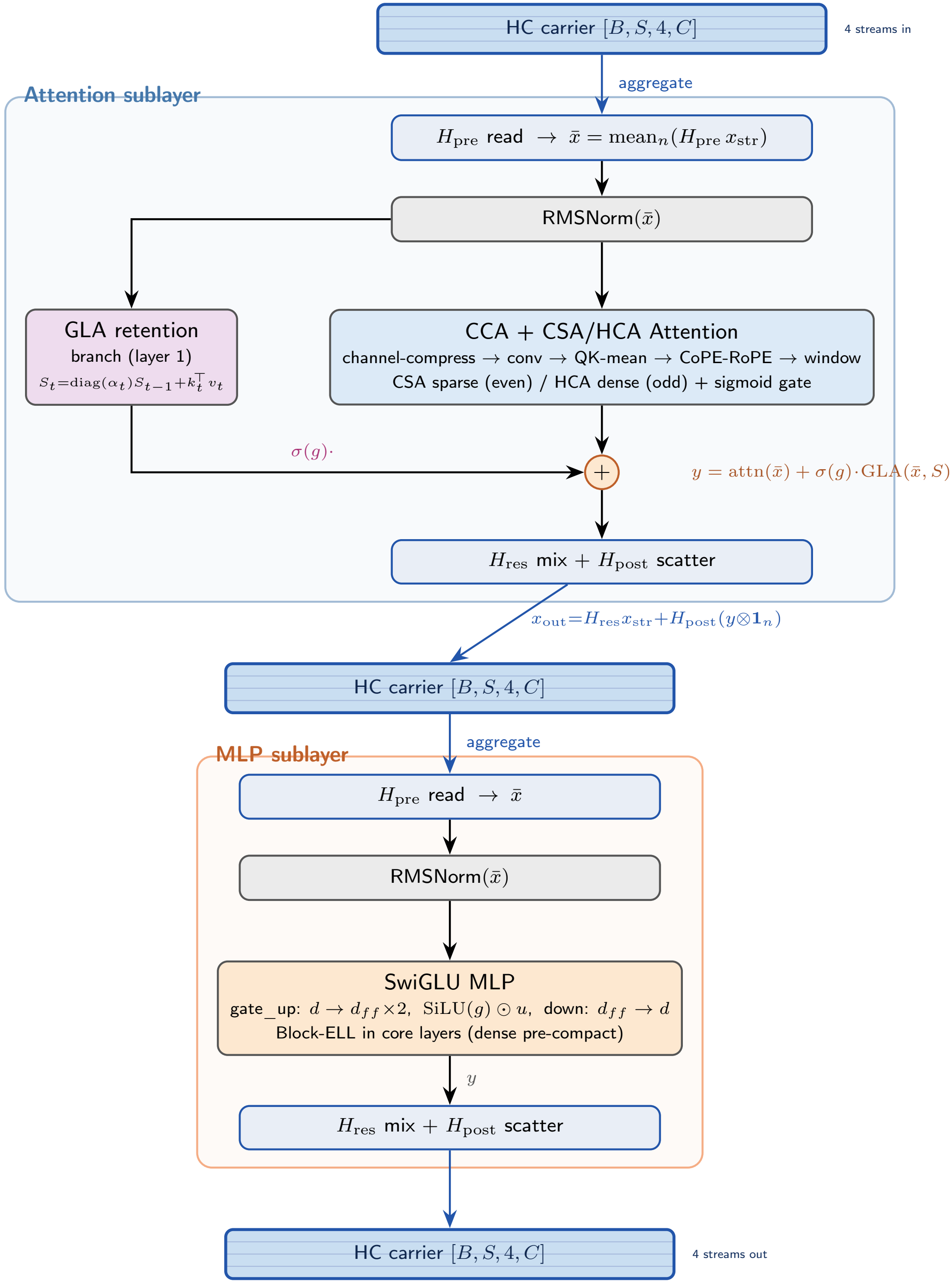
MORPH Block

one block under the Cayley Hyper-Connection carrier (deploy default)



HC carrier (Cayley)

The carrier is $n{=}4$ parallel C-dim streams. Each sublayer:

1. H_{pre} aggregates streams $\rightarrow$ one C-dim $\bar{x}$ that F sees;
2. F (attention or MLP) computes y;
3. H_{res} (orthogonal, Cayley) mixes streams + H_{post} scatters y.

$H_{\text{pre}}, H_{\text{post}}$ row-stochastic; $H_{\text{res}} = \text{Cayley}(\text{skew}) \in O(n)$. Init $\approx x + F(\text{mean}(x))$ (plain residual).

Alternative residual (`residual_mode=null`): **MRR** — split C into compute/context/memory and apply per-channel learned gains γ_i ($\gamma \approx 1/0.5/0.1$). Single-stream $[B,S,C]$; ablation arm, not deploy default.

GLA branch (layer 1 only): runs *parallel* to attention on the same $\bar{x}$; gated sum $y = \text{attn} + \sigma(g)\cdot\text{GLA}$. In core, its state S is carried across loop iterations. $\sigma(g) \approx 0$ at init $\Rightarrow$ off.

Channel slices (compute 384 / context 256 / memory 128) persist under HC as *injection & diagonal-decay targets* within each stream's C — not the residual mechanism. The single-stream injection term broadcasts into all n streams.